

POST-SOFTMAX RANK EXPANSION: A COUNTEREXAMPLE IN TRANSFORMER ANALYSIS

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ABSTRACT. It is commonly assumed in transformer analysis that the softmax operation preserves or reduces the rank of a matrix. We show this assumption is false by constructing an explicit family of rank-1 matrices whose row-wise softmax has full rank. The simplest counterexample is a 2×2 matrix $Z = \begin{pmatrix} 0 & 0 \\ 0 & ab \end{pmatrix}$ with rank 1, whose softmax has rank 2. We generalize to $m \times n$ dimensions via a Vandermonde construction and discuss the implications for low-rank attention approximations.

1. INTRODUCTION

Transformer architectures rely on the attention mechanism [?]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

Researchers frequently analyze the rank of pre-softmax and post-softmax matrices. An implicit assumption is that the softmax, being an element-wise monotone map, does not increase rank. We show this is false.

Main result. There exist rank-1 matrices whose softmax has rank ≥ 2 (and up to $\min(m, n)$ for suitably constructed $m \times n$ matrices).

2. DEFINITIONS

For a real matrix $Z \in \mathbb{R}^{m \times n}$, the *row-wise softmax* is:

$$\sigma(Z)_{ij} = \frac{\exp(Z_{ij})}{\sum_{k=1}^n \exp(Z_{ik})}.$$

The *numerical rank* with tolerance $\tau > 0$ is:

$$\text{rank}_\tau(M) = \#\{i \mid \sigma_i(M) \geq \tau\},$$

where $\sigma_i(M)$ are the singular values of M in descending order.

3. THE 2×2 COUNTEREXAMPLE

Theorem 3.1. *There exists $Z \in \mathbb{R}^{2 \times 2}$ with $\text{rank}(Z) = 1$ and $\text{rank}(\sigma(Z)) = 2$.*

Proof. Take

$$(1) \quad Z = \begin{pmatrix} 0 & 0 \\ 0 & ab \end{pmatrix} = \begin{pmatrix} 0 \\ a \end{pmatrix} \begin{pmatrix} 0 & b \end{pmatrix}$$

with $a, b \neq 0$. Clearly $\text{rank}(Z) = 1$.

Row-wise softmax gives:

$$\sigma(Z) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{1+e^{ab}} & \frac{e^{ab}}{1+e^{ab}} \end{pmatrix}.$$

Suppose $\alpha \cdot [\frac{1}{2}, \frac{1}{2}] + \beta \cdot [\frac{1}{1+e^{ab}}, \frac{e^{ab}}{1+e^{ab}}] = [0, 0]$. From the first component:

$$\frac{\alpha}{2} + \frac{\beta}{1+e^{ab}} = 0 \implies \beta = -\frac{\alpha(1+e^{ab})}{2}.$$

Substituting into the second component:

$$\frac{\alpha}{2} - \frac{\alpha e^{ab}}{2} = 0 \implies \alpha(1 - e^{ab}) = 0.$$

Since $ab \neq 0$, $e^{ab} \neq 1$, so $\alpha = 0$ and $\beta = 0$. Thus the rows are linearly independent, proving $\text{rank}(\sigma(Z)) = 2$. $\square$

Example 3.2. For $a = 2$, $b = 3$, we have $ab = 6$ and

$$\sigma(Z) = \begin{pmatrix} 0.500000 & 0.500000 \\ 0.002473 & 0.997527 \end{pmatrix}.$$

Numerical SVD gives singular values $\sigma_1 \approx 1.1426$, $\sigma_2 \approx 0.4354$, confirming rank 2 to machine precision.

4. GENERAL $m \times n$ CONSTRUCTION

Theorem 4.1. *For any $m \geq 2$ and $n \geq 2$, there exists $Z \in \mathbb{R}^{m \times n}$ with $\text{rank}(Z) = 1$ and $\text{rank}(\sigma(Z)) \geq 2$. In particular, for $m, n \leq 5$, a Vandermonde construction gives $\text{rank}(\sigma(Z)) = \min(m, n)$.*

Proof. Let $u_i = a \cdot i$ and $v_j = b \cdot j$ for $i = 0, \dots, m-1$, $j = 0, \dots, n-1$ with $a, b \neq 0$, and set $Z = uv^\top$. Then

$$\begin{aligned} \sigma(Z)_{ij} &= \frac{\exp(ab \cdot ij)}{\sum_{k=0}^{n-1} \exp(ab \cdot ik)} \\ &= \frac{1}{S_i} (r^i)^j, \end{aligned}$$

where $r = e^{ab} > 1$ and $S_i = \sum_{k=0}^{n-1} (r^i)^k$.

Each row i is proportional to the geometric progression $[1, r^i, r^{2i}, \dots, r^{(n-1)i}]$. For $m \leq n \leq 5$, these rows form a generalized Vandermonde matrix with distinct nodes r^i . The determinant of any $m \times m$ submatrix is $\prod_{0 \leq i < j < m} (r^j - r^i) \neq 0$, so the rows are linearly independent. Hence $\text{rank}(\sigma(Z)) = m = \min(m, n)$. $\square$

Remark 4.2 (Numerical Limitations). For $m, n > 5$, the exponential values r^{ij} span ranges that exceed double-precision arithmetic. The analytical result still holds in exact arithmetic, but numerical verification requires arbitrary precision or a modified construction.

5. IMPLICATIONS

Low-rank attention approximations. Methods such as Performer [?] and Linformer [?] approximate $\sigma(QK^\top) \approx QK^\top$ by treating the softmax as approximately linear. Our theorem shows this approximation has a *fundamental representational gap*: it cannot reproduce rank expansions that the softmax creates.

Practical consequence. If attention is compressed to rank r *before* the softmax, the softmax of the reconstruction may still have higher rank than r . Correct compression must happen *after* the softmax.

Corollary 5.1. *Any theorem assuming $\text{rank}(\sigma(Z)) \leq \text{rank}(Z)$ is incorrect in general.*

6. OPEN QUESTIONS

- (1) **Exact rank formula.** For $Z = uv^\top$, can the rank of $\sigma(Z)$ be expressed in terms of the multiplicities of components in u and v ?
- (2) **Approximate rank.** How close can $\sigma(Z)$ be to a rank- r matrix when Z has rank r ? Is there an Eckart–Young bound for the softmax map?
- (3) **Arbitrary precision.** Can the Vandermonde construction be verified for all m, n using exact rational or symbolic arithmetic?

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Verified empirically on NumPy with double-precision floating point. CPU-only; no GPU required.

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INDEPENDENT RESEARCH

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